



STRUCTURAL ENGINEERING – OCEAN TRANSPORTATION ANALYSIS FOR MODULES

06 – Derivation of Code Load Combinations

This attachment describes the code load combinations that will be used for analysis. The code load combinations described in this appendix are based on DNVGL-ST-N001 (2016), Marine Operations and Marine Warranty.

Ultimate limit states, ULS: (DNVGL section 5.9.8.2)

ULS-a: $1.3(G) + 1.3(Q) + 1.0(D) + 0.7(E)$

ULS-b: $1.0(G) + 1.0(Q) + 1.0(D) + 1.3(E)$

Serviceability limit states, SLS: (DNVGL section 5.9.6.2)

SLS: $1.0(G) + 1.0(Q) + 1.0(D) + 1.0(E)$

Fatigue limit state, FLS: (DNVGL section 5.9.8.2)

FLS = $1.0(E)$

G = permanent loads (WR shifted module weight, stowage load)

Q = variable functional (not applicable)

D = deformation load (not applicable)

E = environmental load (wind, acceleration due to waves, hog/sag, dead load rotation)

Note that permanent loads “G” and variable functional loads “Q” use the same load factor. Environmental loads “E” use a different load factor.

Note that for SPMT transportation, accelerations (from braking) are considered as variable functional loads.

1.0 HEAD SEA CONDITION

Create DNVGL-ST-N001 load combinations with,

G: shifted dead load (DshipFPU, DshipFSU, DshipAPU, DshipASU)

Q: transportation load (included in dead load)

E: head sea forces (“H1” through “H4”)

E: longitudinal wind force (+/- “WML”), is applied in the same direction as surge

E: vessel deformations (“Bh” and “Bs”)

Ultimate Limit State, ULS-a:

(fore, port)

$1.3(\text{DshipFPU}) + 0.7(H1) + 0.7(WML) + 0.7(Bh)$

$1.3(\text{DshipFPU}) + 0.7(H2) + 0.7(WML) + 0.7(Bh)$

$1.3(\text{DshipFPU}) + 0.7(H1) + 0.7(WML) + 0.7(Bs)$

$1.3(\text{DshipFPU}) + 0.7(H2) + 0.7(WML) + 0.7(Bs)$

(fore, starboard)

$1.3(\text{DshipFSU}) + 0.7(H1) + 0.7(WML) + 0.7(Bh)$

$1.3(\text{DshipFSU}) + 0.7(H2) + 0.7(WML) + 0.7(Bh)$

$1.3(\text{DshipFSU}) + 0.7(H1) + 0.7(WML) + 0.7(Bs)$

$1.3(\text{DshipFSU}) + 0.7(H2) + 0.7(WML) + 0.7(Bs)$

(aft, port)

$1.3(\text{DshipAPU}) + 0.7(H3) - 0.7(WML) + 0.7(Bh)$

(aft, starboard)

$1.3(\text{DshipASU}) + 0.7(H3) - 0.7(WML) + 0.7(Bh)$



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$1.3(\text{DshipAPU}) + 0.7(\text{H4}) - 0.7(\text{WML}) + 0.7(\text{Bh})$	$1.3(\text{DshipASU}) + 0.7(\text{H4}) - 0.7(\text{WML}) + 0.7(\text{Bh})$
$1.3(\text{DshipAPU}) + 0.7(\text{H3}) - 0.7(\text{WML}) + 0.7(\text{Bs})$	$1.3(\text{DshipASU}) + 0.7(\text{H3}) - 0.7(\text{WML}) + 0.7(\text{Bs})$
$1.3(\text{DshipAPU}) + 0.7(\text{H4}) - 0.7(\text{WML}) + 0.7(\text{Bs})$	$1.3(\text{DshipASU}) + 0.7(\text{H4}) - 0.7(\text{WML}) + 0.7(\text{Bs})$

Ultimate Limit State, ULS-b:

(fore, port)	(fore, starboard)
$1.0(\text{DshipFPU}) + 1.3(\text{H1}) + 1.3(\text{WML}) + 1.3(\text{Bh})$	$1.0(\text{DshipFSU}) + 1.3(\text{H1}) + 1.3(\text{WML}) + 1.3(\text{Bh})$
$1.0(\text{DshipFPU}) + 1.3(\text{H2}) + 1.3(\text{WML}) + 1.3(\text{Bh})$	$1.0(\text{DshipFSU}) + 1.3(\text{H2}) + 1.3(\text{WML}) + 1.3(\text{Bh})$
$1.0(\text{DshipFPU}) + 1.3(\text{H1}) + 1.3(\text{WML}) + 1.3(\text{Bs})$	$1.0(\text{DshipFSU}) + 1.3(\text{H1}) + 1.3(\text{WML}) + 1.3(\text{Bs})$
$1.0(\text{DshipFPU}) + 1.3(\text{H2}) + 1.3(\text{WML}) + 1.3(\text{Bs})$	$1.0(\text{DshipFSU}) + 1.3(\text{H2}) + 1.3(\text{WML}) + 1.3(\text{Bs})$

(aft, port)	(aft, starboard)
$1.0(\text{DshipAPU}) + 1.3(\text{H3}) - 1.3(\text{WML}) + 1.3(\text{Bh})$	$1.0(\text{DshipASU}) + 1.3(\text{H3}) - 1.3(\text{WML}) + 1.3(\text{Bh})$
$1.0(\text{DshipAPU}) + 1.3(\text{H4}) - 1.3(\text{WML}) + 1.3(\text{Bh})$	$1.0(\text{DshipASU}) + 1.3(\text{H4}) - 1.3(\text{WML}) + 1.3(\text{Bh})$
$1.0(\text{DshipAPU}) + 1.3(\text{H3}) - 1.3(\text{WML}) + 1.3(\text{Bs})$	$1.0(\text{DshipASU}) + 1.3(\text{H3}) - 1.3(\text{WML}) + 1.3(\text{Bs})$
$1.0(\text{DshipAPU}) + 1.3(\text{H4}) - 1.3(\text{WML}) + 1.3(\text{Bs})$	$1.0(\text{DshipASU}) + 1.3(\text{H4}) - 1.3(\text{WML}) + 1.3(\text{Bs})$

Serviceability Limit State, SLS:

(fore, port)	(fore, starboard)
$(\text{DshipFPU}) + (\text{H1}) + (\text{WML}) + (\text{Bh})$	$(\text{DshipFSU}) + (\text{H1}) + (\text{WML}) + (\text{Bh})$
$(\text{DshipFPU}) + (\text{H2}) + (\text{WML}) + (\text{Bh})$	$(\text{DshipFSU}) + (\text{H2}) + (\text{WML}) + (\text{Bh})$
$(\text{DshipFPU}) + (\text{H1}) + (\text{WML}) + (\text{Bs})$	$(\text{DshipFSU}) + (\text{H1}) + (\text{WML}) + (\text{Bs})$
$(\text{DshipFPU}) + (\text{H2}) + (\text{WML}) + (\text{Bs})$	$(\text{DshipFSU}) + (\text{H2}) + (\text{WML}) + (\text{Bs})$
(aft, port)	(aft, starboard)
$(\text{DshipAPU}) + (\text{H3}) - (\text{WML}) + (\text{Bh})$	$(\text{DshipASU}) + (\text{H3}) - (\text{WML}) + (\text{Bh})$
$(\text{DshipAPU}) + (\text{H4}) - (\text{WML}) + (\text{Bh})$	$(\text{DshipASU}) + (\text{H4}) - (\text{WML}) + (\text{Bh})$
$(\text{DshipAPU}) + (\text{H3}) - (\text{WML}) + (\text{Bs})$	$(\text{DshipASU}) + (\text{H3}) - (\text{WML}) + (\text{Bs})$
$(\text{DshipAPU}) + (\text{H4}) - (\text{WML}) + (\text{Bs})$	$(\text{DshipASU}) + (\text{H4}) - (\text{WML}) + (\text{Bs})$

2.0 BEAM SEA CONDITION:

Create DNVGL-ST-N001 unfactored load combinations with,

G: shifted dead load (DshipFPU, DshipFSU, DshipAPU, DshipASU)

Q: transportation load (include with “Dship”)

E: beam sea accelerations (“B1” through “B8”)

E: transverse wind force (+/- “WMT”), is applied in the same direction as sway

Ultimate Limit State, ULS-a:

(fore, port)	(fore, starboard)
$1.3 (\text{DshipFPU}) + 0.7 (\text{B1}) + 0.7 (\text{WMT})$	$1.3 (\text{DshipFSU}) + 0.7 (\text{B5}) - 0.7 (\text{WMT})$
$1.3 (\text{DshipFPU}) + 0.7 (\text{B2}) + 0.7 (\text{WMT})$	$1.3 (\text{DshipFSU}) + 0.7 (\text{B6}) - 0.7 (\text{WMT})$



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1.3 (DshipFPU) + 0.7 (B3) + 0.7 (WMT)	1.3 (DshipFSU) + 0.7 (B7) - 0.7 (WMT)
1.3 (DshipFPU) + 0.7 (B4) + 0.7 (WMT)	1.3 (DshipFSU) + 0.7 (B8) - 0.7 (WMT)
(aft, port)	(aft, starboard)
1.3 (DshipAPU) + 0.7 (B1) + 0.7 (WMT)	1.3 (DshipASU) + 0.7 (B5) - 0.7 (WMT)
1.3 (DshipAPU) + 0.7 (B2) + 0.7 (WMT)	1.3 (DshipASU) + 0.7 (B6) - 0.7 (WMT)
1.3 (DshipAPU) + 0.7 (B3) + 0.7 (WMT)	1.3 (DshipASU) + 0.7 (B7) - 0.7 (WMT)
1.3 (DshipAPU) + 0.7 (B4) + 0.7 (WMT)	1.3 (DshipASU) + 0.7 (B8) - 0.7 (WMT)

Ultimate Limit State, ULS-b:

(fore, port)	(fore, starboard)
1.0 (DshipFPU) + 1.3 (B1) + 1.3 (WMT)	1.0 (DshipFSU) + 1.3 (B5) - 1.3 (WMT)
1.0 (DshipFPU) + 1.3 (B2) + 1.3 (WMT)	1.0 (DshipFSU) + 1.3 (B6) - 1.3 (WMT)
1.0 (DshipFPU) + 1.3 (B3) + 1.3 (WMT)	1.0 (DshipFSU) + 1.3 (B7) - 1.3 (WMT)
1.0 (DshipFPU) + 1.3 (B4) + 1.3 (WMT)	1.0 (DshipFSU) + 1.3 (B8) - 1.3 (WMT)
(aft, port)	(aft, starboard)
1.0 (DshipAPU) + 1.3 (B1) + 1.3 (WMT)	1.0 (DshipASU) + 1.3 (B5) - 1.3 (WMT)
1.0 (DshipAPU) + 1.3 (B2) + 1.3 (WMT)	1.0 (DshipASU) + 1.3 (B6) - 1.3 (WMT)
1.0 (DshipAPU) + 1.3 (B3) + 1.3 (WMT)	1.0 (DshipASU) + 1.3 (B7) - 1.3 (WMT)
1.0 (DshipAPU) + 1.3 (B4) + 1.3 (WMT)	1.0 (DshipASU) + 1.3 (B7) - 1.3 (WMT)

Serviceability Limit State, SLS:

(fore, port)	(fore, starboard)
(DshipFPU) + (B1) + (WMT)	(DshipFSU) + (B5) - (WMT)
(DshipFPU) + (B2) + (WMT)	(DshipFSU) + (B6) - (WMT)
(DshipFPU) + (B3) + (WMT)	(DshipFSU) + (B7) - (WMT)
(DshipFPU) + (B4) + (WMT)	(DshipFSU) + (B8) - (WMT)
(aft, port)	(aft, starboard)
(DshipAPU) + (B1) + (WMT)	(DshipASU) + (B5) - (WMT)
(DshipAPU) + (B2) + (WMT)	(DshipASU) + (B6) - (WMT)
(DshipAPU) + (B3) + (WMT)	(DshipASU) + (B7) - (WMT)
(DshipAPU) + (B4) + (WMT)	(DshipASU) + (B8) - (WMT)

3.0 QUARTERING SEA CONDITION:

Create DNVGL-ST-N001 load combinations with,

G: dead load (“Dship” which represents DshipFPU, DshipFSU, DshipAPU, DshipASU)

Q: transportation load (include with “Dship”)

E: beam sea accelerations (“Q01” through “Q16”)

E: longitudinal wind force (+/- “WML”) is applied in the same direction as surge

E: transverse wind force (+/- “WMT”), applied in the same direction as sway



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DNVGL section 11.6.4 states that the quartering condition can use 60% of longitudinal loads and 80% of transverse loads. Thus,

- Longitudinal wind (“WML”) is multiplied by 0.6
- Transverse wind (“WMT”) is multiplied by 0.8
- Vertical load (“Dship”) is not modified.

Ultimate Limit State, ULS-a:

(fore, port)

1.3(DshipFPU) + 0.7(Q1) + 0.7(+0.6WML + 0.8WMT)
1.3(DshipFPU) + 0.7(Q2) + 0.7(+0.6WML + 0.8WMT)
1.3(DshipFPU) + 0.7(Q3) + 0.7(+0.6WML + 0.8WMT)
1.3(DshipFPU) + 0.7(Q4) + 0.7(+0.6WML + 0.8WMT)

(aft, port)

1.3(DshipAPU) + 0.7(Q09) + 0.7(-0.6WML + 0.8WMT)
1.3(DshipAPU) + 0.7(Q10) + 0.7(-0.6WML + 0.8WMT)
1.3(DshipAPU) + 0.7(Q11) + 0.7(-0.6WML + 0.8WMT)
1.3(DshipAPU) + 0.7(Q12) + 0.7(-0.6WML + 0.8WMT)

(fore, starboard)

1.3(DshipFSU) + 0.7(Q5) + 0.7(+0.6WML -
1.3(DshipFSU) + 0.7(Q6) + 0.7(+0.6WML -
1.3(DshipFSU) + 0.7(Q7) + 0.7(+0.6WML -
1.3(DshipFSU) + 0.7(Q8) + 0.7(+0.6WML -

(aft, starboard)

1.3(DshipASU) + 0.7(Q13) + 0.7(-0.6WML -
1.3(DshipASU) + 0.7(Q14) + 0.7(-0.6WML -
1.3(DshipASU) + 0.7(Q15) + 0.7(-0.6WML -
1.3(DshipASU) + 0.7(Q16) + 0.7(-0.6WML -



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Ultimate Limit State, ULS-b:

(fore, port)	(fore, starboard)
$1.0(\text{DshipFPU}) + 1.3(Q1) + 1.3(+0.6\text{WML} + 0.8\text{WMT})$	$1.0(\text{DshipFSU}) + 1.3(Q5) + 1.3(+0.6\text{WML} -$
$0.8\text{WMT})$	$1.0(\text{DshipFSU}) + 1.3(Q6) + 1.3(+0.6\text{WML} -$
$1.0(\text{DshipFPU}) + 1.3(Q2) + 1.3(+0.6\text{WML} + 0.8\text{WMT})$	$1.0(\text{DshipFSU}) + 1.3(Q7) + 1.3(+0.6\text{WML} -$
$0.8\text{WMT})$	$1.0(\text{DshipFSU}) + 1.3(Q8) + 1.3(+0.6\text{WML} -$
$1.0(\text{DshipFPU}) + 1.3(Q3) + 1.3(+0.6\text{WML} + 0.8\text{WMT})$	
$0.8\text{WMT})$	
$1.0(\text{DshipFPU}) + 1.3(Q4) + 1.3(+0.6\text{WML} + 0.8\text{WMT})$	
$0.8\text{WMT})$	
(aft, port)	(aft, starboard)
$1.0(\text{DshipAPU}) + 1.3(Q9) + 1.3(-0.6\text{WML} + 0.8\text{WMT})$	$1.0(\text{DshipASU}) + 1.3(Q13) + 1.3(-0.6\text{WML} -$
$0.8\text{WMT})$	$1.0(\text{DshipASU}) + 1.3(Q14) + 1.3(-0.6\text{WML} -$
$1.0(\text{DshipAPU}) + 1.3(Q10) + 1.3(-0.6\text{WML} + 0.8\text{WMT})$	$1.0(\text{DshipASU}) + 1.3(Q15) + 1.3(-0.6\text{WML} -$
$0.8\text{WMT})$	$1.0(\text{DshipASU}) + 1.3(Q16) + 1.3(-0.6\text{WML} -$
$1.0(\text{DshipAPU}) + 1.3(Q11) + 1.3(-0.6\text{WML} + 0.8\text{WMT})$	
$0.8\text{WMT})$	
$1.0(\text{DshipAPU}) + 1.3(Q12) + 1.3(-0.6\text{WML} + 0.8\text{WMT})$	
$0.8\text{WMT})$	

Serviceability Limit State, SLS:

(fore, port)	(fore, starboard)
$(\text{DshipFPU}) + (Q1) + 0.6(\text{WML}) + 0.8(\text{WMT})$	$(\text{DshipFSU}) + (Q5) + 0.6(\text{WML}) - 0.8(\text{WMT})$
$(\text{DshipFPU}) + (Q2) + 0.6(\text{WML}) + 0.8(\text{WMT})$	$(\text{DshipFSU}) + (Q6) + 0.6(\text{WML}) - 0.8(\text{WMT})$
$(\text{DshipFPU}) + (Q3) + 0.6(\text{WML}) + 0.8(\text{WMT})$	$(\text{DshipFSU}) + (Q7) + 0.6(\text{WML}) - 0.8(\text{WMT})$
$(\text{DshipFPU}) + (Q4) + 0.6(\text{WML}) + 0.8(\text{WMT})$	$(\text{DshipFSU}) + (Q8) + 0.6(\text{WML}) - 0.8(\text{WMT})$
(aft, port)	(aft, starboard)
$(\text{DshipAPU}) + (Q9) - 0.6(\text{WML}) + 0.8(\text{WMT})$	$(\text{DshipASU}) + (Q13) - 0.6(\text{WML}) - 0.8(\text{WMT})$
$(\text{DshipAPU}) + (Q10) - 0.6(\text{WML}) + 0.8(\text{WMT})$	$(\text{DshipASU}) + (Q14) - 0.6(\text{WML}) - 0.8(\text{WMT})$
$(\text{DshipAPU}) + (Q11) - 0.6(\text{WML}) + 0.8(\text{WMT})$	$(\text{DshipASU}) + (Q15) - 0.6(\text{WML}) - 0.8(\text{WMT})$
$(\text{DshipAPU}) + (Q12) - 0.6(\text{WML}) + 0.8(\text{WMT})$	$(\text{DshipASU}) + (Q16) - 0.6(\text{WML}) - 0.8(\text{WMT})$

4.0 CODE COMBINATIONS BASED ON ISO 19902 (2020)

For reference purposes, ISO code combinations are:

Ultimate limit states, ULS: (Equation 8.2-1)

ULS1: $1.3(\text{GT}) + 1.3(\text{QT}) + 1.0(\text{T})$

ULS2: $1.0(\text{GT}) + 1.0(\text{QT}) + 1.35(\text{T})$

ULS3: $0.95(\text{GT}) + 0.9(\text{QT}) + 1.35(\text{T})$